

MASTER'S THESIS

Towards Destroying the Ring in the Fires of Mount Doom

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Abstract

Placeholder abstract.

Acknowledgements

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Contents

List of Figures	v
List of Tables	vi
List of Notations	vii
List of Abbreviations	viii
1 Introduction	1
1.1 Motivation	1
1.2 Research question	1
1.3 Contributions	2
1.4 Outline	2
2 Background and Related Work	3
2.1 Citing	3
2.2 Figures	4
2.2.1 Colour	6
2.2.2 Sub-figures	7
3 Methodology	8
3.1 Mathematical notation	8
3.2 Equations	8
3.3 Definitions and theorems	8
3.4 Model architecture	9
3.5 Algorithms and code	9

Contents

4 Experiments and Results	12
4.1 Experimental setup	12
4.2 Results	12
4.3 Discussion	12
5 Conclusion	14
Bibliography	15
A Supplementary Material	16
A.1 Additional figures	16

List of Figures

2.1	A fully-connected feed-forward network	6
2.2	A single unit and a recurrent cell	7
3.1	Multi-head attention	9

List of Tables

4.1	Heights of selected characters	13
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List of Notations

$ \cdot $	Measure of magnitude or size: absolute value for real numbers, norm for vectors, cardinality for sets.
d	Dimensionality of a vector.
η	Learning rate.
h	Hidden state vector.
$\mathcal{L}$	Loss function.
n, m	Number of data points, samples or observations.
$\mathbb{N}, \mathbb{R}$	Sets of natural and of real numbers.
p	Probability.
q, k, v	Query, key and value in the attention calculation (Vaswani et al. 2017).
θ	The parameters of a model, collectively.
$w, \mathbf{w}, W$	Weights, for instance those of a neural network.
$x, \mathbf{x}, X$	Input features, or independent variable.
$y, \mathbf{y}, Y$	Target, or dependent variable.
$\hat{y}, \hat{\mathbf{y}}$	Predicted value of the target.

List of Abbreviations

API	Application programming interface
CPU	Central processing unit
GPU	Graphics processing unit
GRU	Gated recurrent unit
LLM	Large language model
LSTM	Long short-term memory
MLP	Multilayer perceptron
NER	Named entity recognition
NLP	Natural language processing
RNN	Recurrent neural network

Chapter 1

Introduction

The number of chapters in this template, and the names they carry, are a suggestion rather than a requirement. Rename them, merge them, split them or drop them as the work demands. What follows is a common arrangement, not a prescribed one.

The same goes for spelling. British, Oxford and American English are all acceptable; what is not acceptable is using two of them in one thesis, which is the usual outcome of never having decided. Pick one before you start writing, and set your spell checker to match. The template is set to British via babel; if you choose American, change that option, since babel also decides hyphenation and the words it generates for you.

1.1 Motivation

Placeholder prose. Replace this section following the four-question structure in the comments above.

1.2 Research question

This thesis addresses the following research question:

One sentence, ending in a question mark, that the rest of the thesis exists to answer. Where the question needs it, one or two further sentences fixing its scope: the setting it applies to, or what a term in it is taken to mean. No more than a short paragraph in total.

1.3 Contributions

It is approached through two sub-questions:

1. A narrower question whose answer the main question depends on.
2. A second one.

1.3 Contributions

This thesis makes the following contributions:

1. A concrete, verifiable claim about something you built or showed.
2. A second one. Three to five items is the usual range.
3. A third one.

1.4 Outline

The front matter carries four reference lists, and they divide into two kinds. The *List of Figures* and *List of Tables* are indexes: they let a reader who remembers a diagram find it again, and they are generated automatically from the short form of each caption, so they need no maintenance beyond writing good captions. The *List of Notations* and *List of Abbreviations* are glossaries: they let a reader who has forgotten what $\mathcal{L}$ or `NER` means look it up without hunting back through the text for its first use. Those two are written by hand in `text/notations.tex` and `text/abbreviations.tex`, and nothing checks them, so a symbol renamed in a later chapter goes stale in the list silently. Keep them in step as you write rather than at the end, and delete either list if the thesis has no notations or abbreviations. Neither replaces introducing a symbol or expanding an abbreviation at first use in the prose.

Chapter 2 reviews prior work on . . . Chapter 3 introduces . . . Chapter 4 evaluates . . . Chapter 5 concludes and discusses limitations.

Chapter 2

Background and Related Work

2.1 Citing

Whenever a *claim* is made, it must either be proven in the manuscript or attributed to published work that supports it. An unsupported claim is the most common thing an examiner marks in a draft.

A claim proven in the manuscript still needs a pointer to its proof, and the reader should never have to go looking. Where the evidence sits in the same section, say so in place, e.g., *the derivation below establishes this*. Where it sits anywhere else, cross-reference it: Chapter 4, never *as shown earlier*. Prefer an explicit \Cref to a bare *above* or *below* even within a single section, because floats move: a figure that migrates to the next page turns *the figure above* into a small lie, and nothing warns you when it happens.

The attribution takes one of two forms, and the difference is grammatical. Use \textcite when the authors are the subject of your sentence: as Sifa et al. (2019) show, financial auditors can use machine learning tools to lighten their workload. Use \parencite when the citation is an aside, supporting a claim you make in your own voice: predicting any price movements of assets in financial markets is quite a difficult task for machine learning models (Deußer and Hillebrand 2018).

Cite a specific page or chapter with the optional argument, (Doe and Roe 2020, p. 42), and give several sources at once by separating the keys with commas (Doe and Roe 2020; Doe 2021).

2.2 Figures

Keep the entries themselves short. An entry needs only what identifies the work, i.e., author, title, venue, year, and the venue should be abbreviated: *Proc. NeurIPS*, not *Proceedings of the 30th International Conference on Neural Information Processing Systems*. Drop pages, publisher, address, editors, month and DOI unless a reader would genuinely struggle to find the work without them. Exporters are the problem here: Google Scholar and publisher sites emit every field they hold, and pasting that in unedited gives a bibliography three times longer than it needs to be, in which the useful information is harder to see. Trim on the way in. The `vaswani2017attention` (Vaswani et al. 2017) entry in `bibliography/references.bib` is the model: four fields, and nothing a reader does not need.

2.2 Figures

Figures are *floats*: LaTeX decides where to put them. Written as a plain `\begin{figure}`, with no optional argument, a figure is placed at the top or bottom of a page, or on a page of its own, wherever it fits without tearing a hole in the running text. That default is almost always the right answer, and it is the reason figures are numbered: the reader is sent to them by number, not by position.

Placement can be forced with an optional argument, and this is where most figure trouble starts. `[h]` asks for *here*; `[!h]` asks more insistently, telling LaTeX to ignore its own limits on how much of a page a float may occupy; `[H]`, from the `float` package, nails the figure in place and refuses to move it at all. Each degrades the page when the figure does not fit: `[!h]` commonly defers it to the next page anyway, having first left a short page behind, and `[H]` opens a large area of white space above the figure rather than reflowing anything.

Reserve forced placement for the cases that genuinely need it, such as a run of small diagrams that must be read in the order they appear, or a figure that a sentence physically points at. Everywhere else, leave the option off.

2.2 Figures

Figure 2.1 carries no placement option, and LaTeX has put it wherever it fitted; the sentence you are reading does not care where that was. It follows that the prose should never say *the figure below*: name the figure, and the sentence stays true wherever the figure lands.

Give every caption a short form as well, as `\caption[short]{long}`: the short one is what the List of Figures prints, which keeps that list scannable instead of reproducing three lines of explanation per entry.

Whenever it is possible, a figure should be *vector* artwork with its text embedded as text: TikZ, or a PDF exported correctly from whatever drew it. Such a figure stays sharp at any zoom, prints at the resolution of the press rather than of your screen, and its labels remain selectable, searchable and set in the document's own font. A screenshot or a PNG gives up all four: it is fixed at whatever resolution it was captured, goes soft in print, and its axis labels are pixels that no one can copy or search. Plotting libraries like Matplotlib will save a PDF instead of a PNG on request, which is usually a one-line change.

Not every diagram is worth writing out in TikZ, and for those we can recommend draw.io,¹ a what-you-see-is-what-you-get editor that exports to PDF and so keeps everything above. Drawing a block diagram by dragging boxes is a good deal faster than positioning nodes by hand, and the result is still vector artwork with real text in it.

A figure taken or adapted from elsewhere must say so in its caption, with a citation like any other borrowed material; Figure 2.1 shows an example how to do this.

For neural network architectures in particular, we recommend the NNTikZ² collection (Love 2024) as a starting point.

¹github.com/jgraph/drawio.

²github.com/fraserlove/nttikz.

2.2 Figures

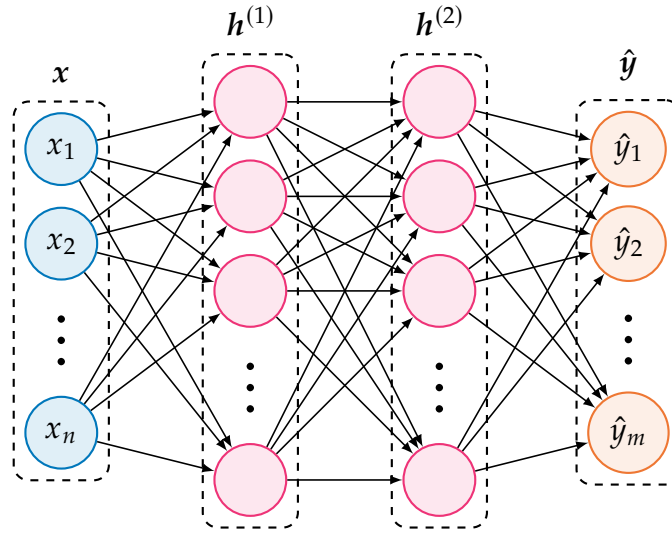


Figure 2.1: A fully-connected feed-forward network mapping an input x to a prediction $\hat{y}$ through two hidden layers. A caption is a full sentence and explains what the reader is looking at. Adapted from Love (2024).

2.2.1 Colour

Whenever colour is used at all, it should come from a colour-blind-safe palette. Red–green colour vision deficiency affects roughly one in twelve men and one in 250 women of European descent, and between 4% and 6.5% of men of East Asian descent (Birch 2012). A figure that distinguishes its series by red against green is therefore unreadable to part of any audience, and a thesis is read by a small audience in which one such reader is enough.

Which palette is a matter of taste: any scheme designed for colour vision deficiency will serve, and several good ones exist. As a starting point we suggest Paul Tol’s schemes,³ of which this template ships the *vibrant* one as a worked example, defined in `aml-thesis.cls` as `tolorange`, `tolblue`, `tolcyan`, `tolmagenta`, `tolred`, `tolteal` and `tolgrey`. Figures 2.1 and 2.2 are coloured this way, showing the palette in action. Both are deliberately more colourful than they likely need to be, in order to display the scheme; colour figures to your own taste.

³Paul Tol, *Colour Schemes*: sronpersonalpages.nl/~pault.

2.2 Figures

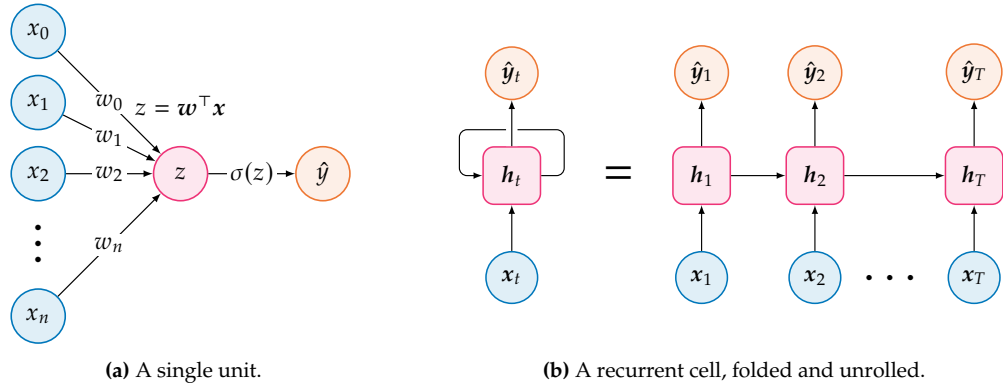


Figure 2.2: Two views of the same machinery: (a) one unit, summing weighted inputs and applying an activation; (b) a recurrent cell shown folded and unrolled over T steps. Each sub-figure takes its own caption and label, and `adjustbox` scales an oversized fragment rather than its coordinates being edited. Adapted from Love (2024).

For figures generated in Python, Tol’s schemes are packaged as `tol_colors`,⁴ which plugs into Matplotlib directly, so plots match the diagrams drawn in TikZ.

2.2.2 Sub-figures

Two panels that belong together should share one figure number rather than take two. Figure 2.2 does this: each panel carries its own caption and label, so Figure 2.2a and Figure 2.2b can be cited separately, while the shared caption explains what the pair has in common.

⁴github.com/Descanonge/tol_colors.

Chapter 3

Methodology

3.1 Mathematical notation

Throughout, scalars are written x , vectors $\mathbf{x}$, matrices $\mathbf{X}$, and the hypothesis space $\mathcal{H} \subseteq \mathbb{R}^d$.

3.2 Equations

The cross-entropy loss over N observations and K classes is defined as (Goodfellow et al. 2016)

$$\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K y_{ik} \log \hat{y}_{ik}(\boldsymbol{\theta}), \quad (3.1)$$

where $\hat{y}_{ik}$ denotes the predicted probability that observation i belongs to class k . Minimising (3.1) with respect to $\boldsymbol{\theta}$ yields the fitted model.

$$z_t = \frac{y_t - \mu}{\sigma}.$$

3.3 Definitions and theorems

Definition 3.1 (Named definition). State the object being defined precisely, and give it a name in the brackets so it can be referred to by name rather

3.4 Model architecture

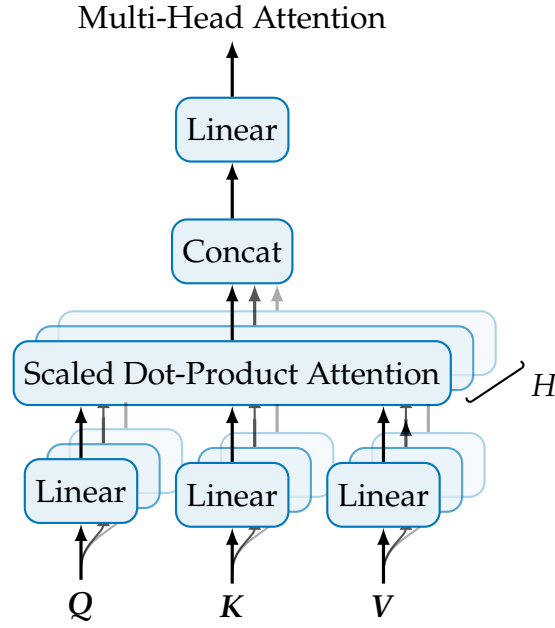


Figure 3.1: Multi-head attention (Vaswani et al. 2017): H heads each project Q , K and V through their own linear map and attend in parallel, and their outputs are concatenated and projected once more. Adapted from Love (2024).

than only by number.

Theorem 3.1. *State the claim. If you prove it, the proof follows in a proof environment, which ends with an automatic QED square.*

Proof. The actual proof. □

3.4 Model architecture

Placeholder description of the model, also shown in Figure 3.1.

3.5 Algorithms and code

A procedure that a reader has to follow step by step belongs in an *algorithm* float rather than in prose. Pseudo-code is numbered, captioned and

Algorithm 1 Perceptron learning algorithm (Rosenblatt 1958)**Input:** Training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ with $x_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$ **Parameters:** Number of epochs T **Output:** Weight vector w and bias b

```

1: Initialize  $w \leftarrow \mathbf{0}, b \leftarrow 0$ 
2: for  $t = 1$  to  $T$  do
3:   for each  $(x_i, y_i) \in \mathcal{D}$  do
4:     if  $y_i(w^\top x_i + b) \leq 0$  then
5:        $w \leftarrow w + y_i x_i$ 
6:        $b \leftarrow b + y_i$ 
7:     end if
8:   end for
9: end for
10: return  $w, b$ 

```

cross-referenced like a figure, and it describes what the method does independently of the language you happened to implement it in: it therefore survives a change of framework, and a reader can follow it without knowing your toolchain. Algorithm 1 gives an example.

State the inputs, parameters and outputs above the body, as there, so that the float can be read without the surrounding text. Keep the body at the level of the argument: name the operations, and leave out the bookkeeping that matters to an implementation but not to the reader. If a step needs justifying, do it in the prose and refer to the line number.

Source code is the other case, and a rarer one. Reproduce it verbatim only when the exact text is the point: an interface a reader will call, a configuration that is hard to describe in words, or a subtlety that the pseudo-code deliberately hides. Listing 3.1 is an example. Keep such listings short, and put anything longer in the appendix or a repository, with a URL in a footnote; a thesis is not the right medium for a file that a reader would rather clone than read.

```

1 def train(model, loader, optimiser):
2     for batch in loader:
3         loss = model.loss(batch)

```

3.5 *Algorithms and code*

```
4     loss.backward()  
5     optimiser.step()  
6     return model
```

Listing 3.1: A training loop, reproduced verbatim. Listings take a caption and a label like any other float.

Chapter 4

Experiments and Results

4.1 Experimental setup

Placeholder description of the setup.

4.2 Results

Table 4.1 reports the main comparison.

4.3 Discussion

Placeholder discussion.

4.3 Discussion

Table 4.1: Heights of selected characters, in centimetres and relative to the tallest. A caption is a full sentence and states the units and the sample, so the table can be read without hunting through the prose.

Character	Height (in cm)	Relative height (in %)
Frodo	124	62.62
Legolas	183	92.42
Gimli	137	69.19
Samwise	127	64.14
Aragorn	198	100.00

Chapter 5

Conclusion

Placeholder summary of what was done and found.

Returning to the research question posed in Section 1.2: placeholder answer, stated plainly and attributed to the evidence that supports it.

Placeholder limitations and future work.

Bibliography

1. Jennifer Birch (2012). 'Worldwide prevalence of red-green color deficiency'. In: *J. of the Optical Society of America A*.
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Appendix A

Supplementary Material

A.1 Additional figures

Placeholder appendix material.